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ITAI 2372
A10: Advantages of using AI in Government and Public Sector

1. Introduction

Artificial intelligence is no longer just a theoretical concept, as it has become a highly practical tool for everyday governance. Governments worldwide are actively using machine learning, natural language processing, and data analytics to modernize their systems, cut costs, and improve the services they provide to citizens and consumers. In fact, the Organization for Economic Co-operation and Development (OECD) has already documented over 200 real-world examples of AI in action across 11 core government functions, ranging from financial management and justice administration to anti-corruption efforts.

Fraud detection stands out as one of the most measurable and impactful uses of AI in the public sector. Because government agencies distribute trillions of dollars each year, they face massive exposure to fraud, waste, and improper payments. Traditional, rule-based monitoring systems simply cannot keep up with the speed and sophistication of modern fraud, a reality that became painfully obvious during the surge of digital payments tied to pandemic-era relief programs.

The United States Department of the Treasury uses machine learning through its Office of Payment Integrity (OPI). It covers the core problem, the specific AI techniques implemented, the impressive outcomes, and the limitations of the current system. Finally, the report proposes a new, overarching AI application designed to further boost public sector integrity and public trust.

2. Case Study Analysis

Case Study: U.S. Department of the Treasury — AI-Powered Fraud Detection (FY2023–FY2024)

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Problem or Need Addressed

As the world's largest disburser of public funds, the U.S. federal government faces a massive security challenge. The Treasury Department's Bureau of the Fiscal Service processes roughly 1.4 billion payments worth over $6.9 trillion every year for more than 100 million people. This immense volume creates a complex and highly vulnerable landscape for fraud.

Fraud skyrocketed during the COVID-19 pandemic, as stimulus checks, expanded unemployment, and small business loans drew the attention of sophisticated criminal networks. By 2023, experts projected global online payment fraud to surpass $362 billion by 2028, and check fraud, whether physical or digital, became a massive issue for the U.S. government specifically. Criminals stole mail and hijacked accounts, rapidly laundering fraudulent checks through complicit banks and money mules.

The biggest problem wasn't just the sheer volume of fraud, but the speed. Fraudulent checks were cashed in hours, making manual reviews and rigid rule-based systems too slow to stop the money from leaving the system. The Treasury's OPI desperately needed a way to monitor billions of transactions in near real time.

2.2 AI Tools and Techniques Implemented

To fight this, the OPI deployed a suite of machine learning tools, deliberately avoiding generative AI like large language models. The main technologies included:

Machine Learning Anomaly Detection: Models were trained on historical transaction data to spot complex fraud patterns. By flagging unusual payee combinations, weird transaction timing, geographic mismatches, and strange check amounts, these models detected anomalies at a scale human reviewers never could.

Risk-Based Screening: A probabilistic scoring system evaluates the risk of every single payment, pausing high-risk transactions for extra review before the money is sent out. This successfully shifted the Treasury from reacting to fraud to actively preventing it.

Big Data Integration: The OPI connected its system with the Do Not Pay (DNP) database, which tracks ineligible recipients, deceased individuals, and other high-risk entities. In May 2024, they partnered with the Department of Labor to share this intelligence with state unemployment agencies.

Real-Time Bank Alerting: When the system spots an anomaly, it immediately alerts the banking institution before the check can be cashed, closing the window for financial loss.

Treasury officials have publicly confirmed that this system does not rely on generative AI. Instead, it uses predictive machine learning and big data analytics; which are approaches that are much easier to explain and have stronger regulatory backing than newer, less predictable models.

2.3 Outcomes and Benefits Achieved

The results of the Treasury’s AI deployment offer a highly documented and compelling benchmark for government AI success:

Outcome Category	**FY2023 Result**	**FY2024 Result**
Total Prevented / Recovered	$652.7 million	$4.0+ billion

Outcome Category	FY2023 Result	FY2024 Result
Check Fraud Recovery (ML)	$375 million	$1 billion
High-Risk Transaction Prevention	—	$2.5 billion
Risk-Based Screening Prevention	—	$500 million
Year-over-Year Growth	Baseline	+513%

Table 1: Treasury OPI AI Fraud Detection Outcomes —FY2023 vs. FY2024

The 513% jump in recovered and prevented funds—from $652.7 million in FY2023 to over $4 billion in FY2024—highlights how quickly the system matured and expanded to new partners. It wasn't just about saving money, either; the AI generated leads that directly resulted in active criminal cases and arrests, proving its value as a true actionable intelligence tool.

Beyond the dollars saved, the program set an excellent standard for agencies working together. The partnership with the Department of Labor showed how state programs can leverage federal machine learning tools without footing the bill for duplicate development.

2.4 Challenges and Limitations Observed

Even with its documented success, the Treasury's AI program—and government AI overall—faces real challenges that we must acknowledge:

Challenge	**Description**
Biased Training Data	Models trained strictly on historical data can miss brand-new fraud types. Michigan's MiDAS welfare fraud algorithm failed 93% of the time because of this exact problem.
AI Feedback Loops	Models that learn from their own outputs can reinforce biases over time, narrowing their focus to known patterns while missing novel schemes.
Data Quality Dependency	The GAO emphasizes that poor-quality training data directly degrades model reliability and destroys public trust in AI systems.
Explainability Deficit	Complex machine learning models produce outputs that are notoriously difficult to explain to judges, auditors, or the general public.

Challenge	Description
Privacy and Civil Liberties	Broad screening raises serious concerns about surveillance and due process, especially when algorithmic flags trigger law enforcement actions.

The Government Accountability Office (GAO) stresses that high-quality data is the foundational backbone of any AI fraud system. A 2025 GAO report warned that feeding bad data into an AI makes the models unreliable, and systemic errors destroy public trust. While the GAO recommended creating a permanent analytic center focused entirely on fraud and improper payments, this has not been fully implemented yet. The stakes here are high: Michigan's flawed MiDAS algorithm generated erroneous predictions 93% of the time, resulting in wrongful debt collections against tens of thousands of innocent citizens.

To understand how this operates in practice, consider a specific case of a widespread check-washing ring intercepted by the Treasury in late 2023. Criminal networks had stolen thousands of physical Treasury-issued checks—primarily tax refunds and social security disbursements—from mailboxes across multiple states. Using chemicals, they "washed" the original payee names and amounts, replaced them with fraudulent information, and deployed money mules to cash them at various regional banks.

Under the old legacy system, these checks would have likely cleared because they looked legitimate on paper and were processed individually. However, the newly implemented AI system detected the scheme by analyzing behavioral and geographic anomalies. The machine learning models flagged an impossible velocity of check clearings occurring in highly specific geographic clusters that completely mismatched the original recipients' home addresses. By cross-referencing these anomalies with financial databases and applying probabilistic risk-scoring, the AI automatically triggered real-time alerts to the banking institutions involved. The banks froze the transactions just before the funds were disbursed, successfully recovering millions of dollars in a single sweep and providing law enforcement with the exact digital footprints needed to arrest the perpetrators.

3. Innovative Proposal: AI-Driven Cross-Program Fraud Intelligence Network (CFIN)

3.1 Proposed Application

I propose creating a Cross-Program Fraud Intelligence Network (CFIN)—a federated, AI-powered platform that shares anonymized fraud signals across all federally funded programs in real time. While the Treasury currently focuses only on its own payment layer, CFIN would act as an overarching intelligence network to spot fraud rings that span multiple agencies and jurisdictions.

The key innovation here is federated machine learning, a method where individual agencies train models on their own local data without actually sharing sensitive personal information. Agencies like the Social Security Administration, Medicare, and the Department of Education would train local models on their respective datasets. A central server would then aggregate only the model weight updates—never the raw data—to create a continuously improving, shared fraud model. This approach catches fraud that no single agency could find alone, while fully preserving citizen privacy under HIPAA and the Privacy Act.

Additionally, graph neural networks could map relationships between IP addresses, bank accounts, and mailing addresses. Because fraud rings often use the same synthetic identities and money mules across multiple programs, GNNs would expose these hidden connections. A criminal who looks completely legitimate in one agency's system would be instantly flagged when their network is analyzed holistically.

3.2 Justification and Expected Outcomes

The reality is that sophisticated criminals don't limit themselves to scamming just one program. Pandemic-era data proved that organized networks exploited unemployment, stimulus checks, and loan programs all at once, often using the same stolen identities. Current, isolated detection systems miss these broad attacks entirely.

CFIN would provide immense benefits, including:

- Detecting cross-program fraud rings currently invisible to siloed agencies.
- Reducing false positive rates by sharing richer context across departments.
- Spotting emerging fraud schemes faster, as a threat to one agency immediately warns all others.
- Generating massive cost savings by sharing AI infrastructure instead of duplicating it.

If the Treasury's single-agency system saved $4 billion in FY2024 alone, a unified federal network combining signals from dozens of programs could conservatively expose ten times that amount in fraud.

3.3 Potential Challenges

Implementing CFIN does come with realistic hurdles:

Legal and Regulatory Complexity: Agencies operate under strict, differing data-sharing laws. CFIN would require new legislation or expanded authority under current OMB directives, and every agency's involvement would need explicit authorization.

Infrastructure Costs: Setting up secure, production-grade federated learning requires serious upfront funding, although the long-term savings from consolidated tech development should outpace those initial costs.

Governance: A cross-agency system means shared accountability gaps. Clear rules for model ownership, bias auditing, and error remediation must be ironed out before deployment.

Adversarial Adaptation: Sophisticated fraudsters will inevitably adapt once they figure out the new AI parameters. CFIN would need continuous retraining and robustness testing to stay effective over time.

Despite these challenges, CFIN is highly feasible. The Treasury's Do Not Pay system and its 2024 Department of Labor partnership already prove that inter-agency AI collaboration is workable within current legal frameworks. CFIN is simply the natural next step. The U.S. Department of the Treasury’s Office of Payment Integrity runs one of the most successful, well-documented AI programs in the government today. By preventing or recovering over $4 billion in FY2024, a massive 513% increase from the previous year, it proves that AI is no longer a theoretical enhancement, but a highly profitable tool for protecting public funds. However, this case study highlights both the power and the immense responsibility of using AI at scale. While machine learning and real-time alerts catch complex fraud that legacy systems miss, issues surrounding biased data, algorithmic transparency, and privacy require constant institutional oversight. The GAO is entirely correct to push for a permanent fraud analytic center of excellence to provide structured governance.

The proposed Cross-Program Fraud Intelligence Network (CFIN) takes the Treasury's success and expands it to cover its primary limitation: its single-agency scope. By utilizing federated machine learning and graph neural networks, agencies could finally expose the widespread fraud networks that easily slip through the cracks today. With the right governance, legal frameworks, and funding, CFIN could save billions more while fiercely protecting citizen privacy and civil liberties.

As digital payments grow and AI capabilities advance rapidly, the question is no longer *whether* the government should use AI to fight fraud, but *how* to deploy it responsibly at a massive scale. The Treasury has built a fantastic foundation; the challenge now is to build upon it.

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